OFFICE OF RESEARCH AND GRADUATE SCHOOL CURATION PROGRAM

A Comparative Study on the Performance of Modern Graph Databases

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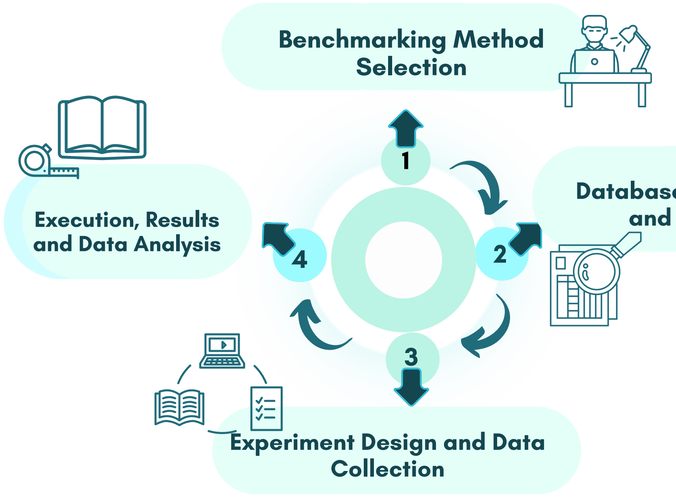
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Research Objective

This research aims to compare and assess the scalability, performance, and applicability of multiple graph databases for diverse kinds of graph data applications.

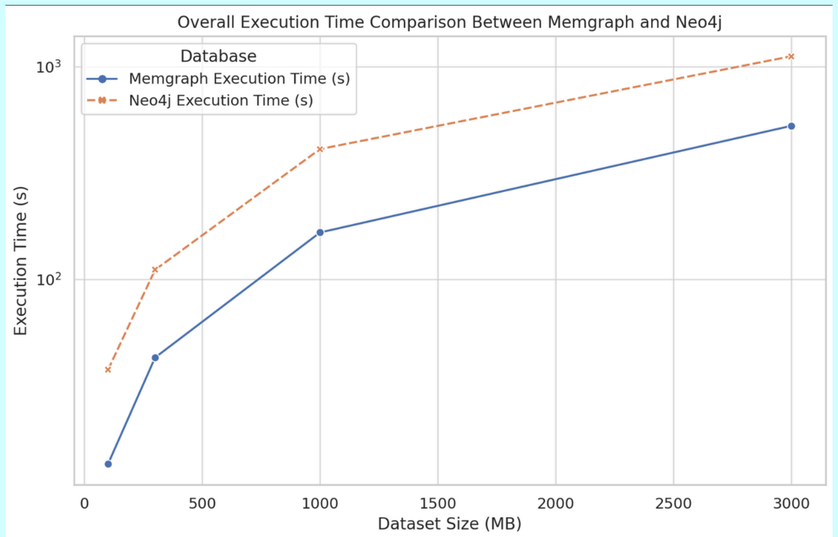
Methodology



Literature Review

- **Comparison of Graph Database Models:**
  - A 2012 review compared AllegroGraph, DEX, HypergraphDB, Neo4j, and others based on their data structures, query languages, and integrity constraints, but without performance evaluation. Our goal was to make experimental evaluation the foundation of our research.
- **Experimental Evaluation of Graph Databases:**
  - Experimental comparisons of graph databases, such as the 2013 and 2023 studies, have provided insights into the performance of different databases under various conditions.

Findings



Memgraph Performed Better than Neo4j for loading data

Discussion

Assessment of Apache HugeGraph and RedisGraph:

- Both Databases lack efficient built-in query methods to load data from CSV files.
- Both Databases lack clear documentation for troubleshooting purposes.
- The Python driver for Apache HugeGraph has non-functioning methods which leads to additional work in debugging
- Apache HugeGraph has a data visualisation tool with poor accessibility

Scalability Analysis Between MemGraph and Neo4j:

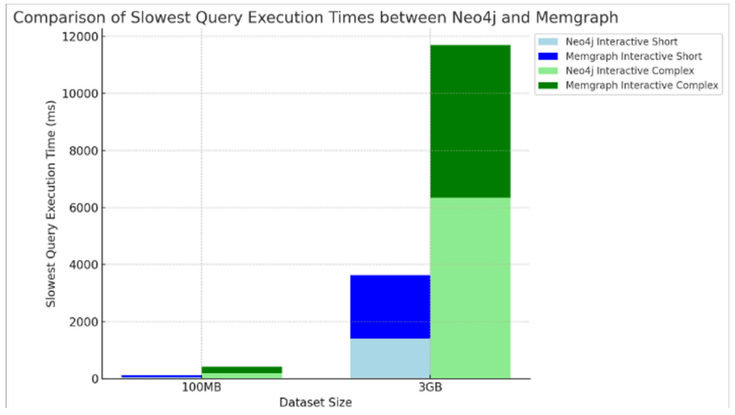
- **Memgraph:** Scales more efficiently, with its loading times increasing at a slower rate relative to data size
- **Neo4j**: Experiences a more significant slowdown as data size increases, particularly with large datasets (1GB and 3GB), where the increase in loading time is much steeper.

Conclusion

Through rigorous benchmarking, we discovered that while some databases excel in handling **large-scale**, complex queries with **remarkable efficiency**, others offer **robust scalability** and **resource optimization**. As the landscape of data management continues to evolve, our findings provide valuable guidance for researchers and practitioners alike.

Figure 1: Feature Comparison Between the Databases

Feature	Memgraph	Neo4j	Apache HugeGraph	RedisGraph
ACID Compliance	✓	✓	✓	✓
Cypher Query Language	✓	✓	✗	✓
Gremlin Query Language	✗	✗	✓	✗
Graph Algorithms	✓	✓	✗	✗
Full-Text Search	✓	✓	✗	✗
Horizontal Scalability	✗	✓	✓	✓
Open Source	✓	✗	✓	✓
High Availability	✓	✓	✓	✓
Graph Visualization Tools	✓	✓	✗	✗
Multi-Model Data Support	✗	✗	✓	✗
Active Community Support	✓	✓	✗	✓
Enterprise Support	✓	✓	✓	✓



References

[1] A. Lopes, D. Rodrigues, J. Saraiva, M. Abbasi, P. Martins, and C. Wanzeller, "Scalability and Performance Evaluation of Graph Database Systems: A Comparative Study of Neo4j, JanusGraph, Memgraph, NebulaGraph, and TigerGraph," 2023 Second International Conference On Smart Technologies For Smart Nation (SmartTechCon), Singapore, Singapore, 2023, pp. 537-542, doi: 10.1109/SmartTechCon57526.2023.10391694.

[2] J. Monteiro, F. Sá, and J. Bernardino, "Experimental Evaluation of Graph Databases: JanusGraph, NebulaGraph, Neo4j, and TigerGraph," Appl. Sci. 2023, 13(9), 5770, doi: 10.3390/app13095770.